SIDDHANT SINGH

Bangalore, India · +91-87478 93867 · siddhant3103@gmail.com · LinkedIn · GitHub · LeetCode

AI/ML engineer (B.Tech CSE, VIT, CGPA 8.99, graduated May 2026) specializing in **making models small and fast enough to run on-device**: knowledge distillation, INT8 quantization-aware training, and LLM quantization. Ex-Samsung R&D research intern. Also ship production GenAI systems (RAG, streaming LLM backends, offline voice AI). LeetCode 700+ solved, contest rating 1837. AWS Certified.

SKILLS

Languages: Python · C++ · Java · C · JavaScript · SQL

ML / DL: PyTorch · TensorFlow / TFLite · HuggingFace · Transformers · scikit-learn · OpenCV · CNNs · CLIP

Model Optimization: Knowledge Distillation · INT8 PTQ & Quantization-Aware Distillation · NF4 / mixed-precision · GGUF / llama.cpp · ONNX Runtime · bitsandbytes · NNAPI delegation

GenAI / LLM Systems: Agentic AI · RAG · Qdrant · Prompt Engineering · LLM APIs (Gemini, Groq) · Whisper (ASR) · LangChain · LangGraph · Vision-Language Models · Structured / JSON-constrained output

Backend & MLOps: FastAPI (async) · SSE streaming · OAuth 2.0 · MLflow · pytest · Docker (basics) · MongoDB · GitHub Actions (CI) · PostgreSQL · SQLite · Redis · Git · Linux

Cloud: AWS Certified Cloud Practitioner · EC2 · SageMaker (basics)

EXPERIENCE

AI/ML Research Intern | Samsung R&D Institute India *Jan 2026 – Aug 2026*

On-device document image classifier · EfficientNet-B4 -> MobileNetV2 · TFLite · Android

- Built a 10-class document classifier for budget/mid-range Android phones: distilled an EfficientNet-B4 teacher (95.1%) into a MobileNetV2 student and quantized to INT8 with **quantization-aware distillation: 97.9% test accuracy (above the teacher), 2.9 MB, under 10 ms inference on NPU** and under 50 ms on CPU-only older devices.
- Diagnosed plain INT8 PTQ collapsing accuracy (F1 0.95 -> 0.74), switched to QAD to recover it. Traced about 100 ms app latency to the NNAPI delegate not engaging (not model capacity); found and fixed a double-preprocessing bug and a uint8 wraparound eval bug.
- Measured the real-world gap: on a 486-image phone-captured OOD set, accuracy fell from 95–98% (clean) to **84.1%**. Traced "book-attractor" misclassification to the teacher's own decision boundary; targeted data collection (819 -> 1,619 images) lifted magazine-cover recall from 3–8% to 54–69%.
- Built a 2-stage dataset-cleaning pipeline (pHash dedup + CLIP zero-shot triage) and shipped an Android app: gallery/camera classification, auto-sorting into class folders, live-camera mode, fully offline.

PROJECTS

Vaani: On-Device Hindi-Hinglish Voice Intelligence

Python · FastAPI · Whisper-small (ONNX INT8) · Phi-3-mini (GGUF INT4) · llama.cpp · MLflow

- Voice-note engine that turns Hinglish audio into summary, action items and sentiment. **Dual-engine design**: one env flag switches between fully **offline CPU inference (no GPU, no cloud**, Rs.0/month, audio stays on device) and a **cloud-API mode powering the deployed Telegram bot** (FastAPI on Render) since the on-device build can't be hosted for free.
- Quantization benchmarks (MLflow + llama-bench): LLM footprint **–68.7% (7.29 -> 2.28 GB)**, generation speed **2.1× (3.70 -> 7.81 tok/s)** on CPU.
- Async FastAPI proxying a native llama-server daemon (no GIL blocking); 30 s sliding-window chunking + forced language hint fixed Whisper hallucination and OOM on long audio; JSON-schema-constrained decoding guarantees parseable output every call.

Prompt Memory: Full-Stack GenAI Application *Live on Vercel*

FastAPI · Qdrant · MongoDB · Groq LLM + Whisper · OAuth 2.0 · SSE

- Production RAG-style backend: 5-stage LLM pipeline (retrieval -> semantic enrichment -> prompt expansion -> LLM -> streamed response) over Qdrant and multilingual Sentence-Transformers, giving 4×–110× context expansion.
- Voice-to-prompt via Groq Whisper (20+ languages incl. Hinglish); multi-key API rotation with retry and cooldown tracking, so no user-facing rate-limit errors. pytest suites mock LLM / vector-DB calls to test retries and streaming, run automatically by GitHub Actions on every push.

AQCAP: Adaptive Quantization & Context-Aware Pruning (Capstone)

PyTorch · HuggingFace · bitsandbytes · Phi-3-mini

- Profiled all 32 layers of Phi-3-mini (3.82B) via PyTorch hooks to derive a Layer Sensitivity Index guiding NF4/INT8 mixed precision: **VRAM –58.3% (7.29 -> 3.04 GB)** for code-vulnerability detection under a 4 GB budget.
- F1 0.85 / Recall 0.93 at zero training cost, beating a supervised baseline (F1 0.83) across 6 configurations; prompt engineering (+0.10 F1) proved a stronger lever than architecture changes.

EDUCATION

B.Tech, Computer Science & Engineering | Vellore Institute of Technology *Graduated May 2026 · CGPA 8.99 / 10*

ACHIEVEMENTS

- **LeetCode: 700+ problems solved, contest rating 1837. VIT Top Coders: top 200 of 15,000** students in competitive programming.
- **GATE 2026: AIR 3966. AWS Certified Cloud Practitioner**.